

Emma Ricci-DeLucca

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I aim to understand the gap between what AI can predict from wearable data and what care providers can act on, with the goal of designing more effective human-AI collaboration in healthcare.

RESEARCH IN PROGRESS (2026)

DoulaConnect: AI-Enabled Health Platform

- 1st place among 8 teams and 40 participants at the 2026 Dartmouth Innovation Accelerator for Digital Health pitch competition (\$40,000 award), judged by top investors and executives
- Preliminary results indicate DoulaConnect is feasible and acceptable, with a medium favorable effect on postpartum depression symptoms in women in rural areas
- Co-founder, leading front-end UX/UI co-design and testing of an AI-enabled platform with 60+ stakeholders (administrators, clinicians, doulas, patients)

Wearable Sensing ML Pipeline

- Built Random Forest classifiers achieving a mean AUC of 0.916 for individual-level depression detection, using SHAP analysis to identify daily activity level as the strongest predictive signal
- Designed an ML pipeline combining wearable Fitbit sensor data (heart rate, sleep, activity) with clinical records (EHR) to detect postpartum depression at scale
- Applied statistical modeling (regression, mixed-effects models) to identify which wearable signals differ significantly between depressed and non-depressed patients

Human-Centered Qualitative User Research

- Led 14 semi-structured interviews with pregnant and postpartum individuals, investigating symptom recognition, care-seeking behavior, and provider communication in the perinatal period
- Applied inductive thematic analysis (Atlas.ti), iteratively coding transcripts to surface themes and synthesized findings into design implications for sensing and intervention systems

PROFESSIONAL EXPERIENCE

NSF Innovation PhD Trainee, Dartmouth College

Sep 2023 - Present

- Selected for Innovation Program, a selective fellowship supporting translation of dissertation research into a commercial venture through funded entrepreneurship training and startup support

Global Business Development Intern, Siemens Healthineers

Feb 2022 – Nov 2022

- Managed Agile portfolios for 9 products in Azure DevOps, modeled 5 business plans, and built 2 Excel dashboards to support SaaS customer acquisition strategy and expansion into 15 new markets

Digital Health Design Fellow, Duke University

Aug 2021 – May 2023

- 1 of 20 fellows chosen from 91 applicants to lead ethnographic research with clinicians, translating insights into prototypes and a GTM strategy to reduce overcrowding and improve patient safety

Additive Manufacturing Researcher, Duke University

Jun 2020 – Jan 2022

- Applied FEA to porous, metallic orthopedic implants, improving understanding of their mechanical behavior and informing future implant design
- Wrote a systematic literature review synthesizing findings across 181 studies, screened from 341 total, to identify evidence gaps in implant performance

Computational Microscopy Researcher, Swarthmore College

Jun 2020 – Aug 2020

- Awarded Lang Fellowship to build convolutional neural network (TensorFlow) for reconstructing high-resolution 3D images from low-resolution inputs

Mechanobiology Engineering Researcher, University of Pennsylvania

May 2019 – Aug 2019

- 1 of 18 chosen from 130+ applicants for NSF research to investigate tumor growth and metastasis mechanics. Co-authored a peer-reviewed publication and presented findings at BMES 2019

EDUCATION

Dartmouth College

PhD Candidate, Human Computer Interaction, *GPA: 3.9*

2023 – Present

Duke University

MEng in Biomedical Engineering & Medical Device Design, *GPA: 3.8*

2021 – 2023

Swarthmore College

BS Engineering, *GPA: 3.7*

2017 – 2021

PUBLICATIONS

"I Just Need My Skin to Be Free": Understanding Consumer Wearable Experiences and Design Opportunities for Perinatal Mental Health Monitoring

Ricci-DeLucca, Batool, Jarsve, Moore, Heavey, & Murnane

Under Review, IMWUT (Interactive, Mobile, Wearable and Ubiquitous Technologies)

Digital Phenotyping of Perinatal Mental Health: A Systematic Review of Machine Learning and Wearable Sensor Approaches

Ricci-DeLucca, Heavey, & Murnane

2026, Journal of Affective Disorders Reports

Family In The Loop: Enabling Family Involvement in Dementia Care at Long-term Facilities with Person-centered AI Tools

Moore, Tao, **Ricci-DeLucca**, Tadin, Tadin, Morgan, Nottelmann, & Murnane

2025, CSCW (Computer-Supported Cooperative Work)

Wearable Biosensors for Early Detection and Continuous Monitoring of Postpartum Depression: A Human-Centered Approach

Ricci-DeLucca, & Murnane

2024, CHI (Conference on Human Factors in Computing Systems) Workshop Paper

Review of In-Vivo Behavior of Porous Titanium Implants for Orthopedic Applications

Ricci-DeLucca, Kelly, & Gall

2023, Journal of Medical Implants and Surgery

Gaussian curvature dilutes the nuclear lamina, favoring nuclear rupture, especially at high strain rate
Pfeifer, Tobin, Cho, Vashisth, Dooling, Vazquez, **Ricci-De Lucca**, Simon, & Discher

2022, Nucleus

SKILLS

Research & UX: Mixed-Methods, Qualitative Research (Interviews, Surveys, Focus Groups), Thematic Analysis, Quantitative Empirical Study, Usability Testing, Participatory Design, Ethnographic Research

AI & ML: scikit-learn, PyTorch, Wearable Sensing & Prediction, Human-Centered ML/NLP, Data Analysis

Programming: Python, R, HTML, d3.js, Pandas, C/C++, NumPy

Design: Figma, Adobe Creative Cloud, Canva, Rapid Prototyping

Languages: English (native), Italian (native), French (fluent)

MENTORSHIP

Graduate Student Mentor, Dartmouth College

Sep 2024 - Present

- Mentored 7 undergraduate students on maternal and perinatal mental health research, recognized with Outstanding Graduate Research Mentorship Award
- Secured a combined \$40,000 in competitive research funding across multiple scholarships and fellowships to fund undergraduate researchers

HONORS & AWARDS

Outstanding Graduate Student Mentor, Dartmouth College

2026

Best Rural Healthcare Innovation, New England MedTech Conference

2025

Graduate Student Council Student Award, Dartmouth College

2025

Grad Cohort for Women, Computing Research Association

2025

Travel Grant, Neukom Institute for Computational Science

2024

SERVICE

Student Volunteer

CHI 2026 | FemTechnology Summit 2023

Evaluator for Peer-Reviewed Journals

IMWUT 2026 | CHI 2026 | CHI 2024

Mental Health & Wellness Student Leader, Dartmouth College

2023 - 2025

Animal Welfare Volunteer & Advocate, PA, NC, NH & Italy

2020 - Present